

6

Future Trends

6.1 Ubiquitous Computing

Q.1 Write short note on ubiquitous computing.

Ans. :

- Ubiquitous computing (ubicom) is a post-desktop model of human-computer interaction in which information processing has been thoroughly integrated into everyday objects and activities.
- More formally Ubiquitous computing is defined as "machines that fit the human environment instead of forcing humans to enter theirs."
- All models of ubiquitous computing share a vision of small, inexpensive, robust networked processing devices, distributed at all scales throughout everyday life and generally turned to distinctly common-place ends.
- For example, a domestic ubiquitous computing environment might interconnect lighting and environmental controls with personal biometric monitors woven into clothing so that illumination and heating conditions in a room might be modulated, continuously and imperceptibly.
- Context-aware/awareness originated as a term from ubiquitous computing or pervasive computing which sought to deal with linking changes in the environment with computer systems.

Context aware systems are concerned with 3A :

1. Acquisition of context (e.g. using sensors to perceive a situation),
 2. Abstraction and understanding of context (e.g. matching a perceived sensory stimulus to a context) and
 3. Application based on the recognized context (e.g. triggering actions based on context)
- Active context awareness : "An application automatically adapts to discovered context by changing the application's behavior"
 - Passive context awareness : "An application presents the new or updated context to an interested user or makes the context persistent for the user to retrieve later"
 - Contextual information is a set of data, gathered from sensors, applications and users, that conforms to a context model and provides a snapshot that approximates the real-world context at a given point in time.
 - Context-aware computing issues :
 1. Context is most useful in dynamic, mobile environments. But what is the relevant information in various situations ?
 2. Mobility results in continuous updates of context information. How can we efficiently manage this ?
 3. How can we share context ?
 4. How do we handle uncertainty of context information ?
 5. How do we ensure privacy control and management of context information ?
 6. How do we reach a common understanding of implications and semantics of (shared) context information ?
 7. How to effectively use context with resource restrictions ?
 8. How to exploit the past context or context history ?

Q.2 What is an implicit interaction ?**Ans. :**

- Observing communication between humans, we can see that a lot of information is only exchanged implicitly. The way people interact with each other and also the situation in which they interact carries information that is often implicitly exploited in the exchange of messages.
- While in a conversation the behaviour of participants as well as what happens in the surrounding environment supplies valuable information that is often vital for the understanding of messages.
- In many cases the robustness of human-to-human communication is based on the implicitly introduced contextual information, such as gestures, body language and voice.
- Another example is redundancy between body language (e.g. nodding) and spoken languages (e.g. the word 'yes'). This implicitly introduced knowledge is also used to disambiguate information. e.g. in a discussion with a student pointing at a computer the term 'sun' has a different meaning than the same term when on the beach together with friends; a more in depth discussion is given in.

Q.3 Explain case study of Ambient Wood.**Ans. :**

- The Ambient Wood project, carried out as part of the equator project, set out to provide an augmented learning experience for children in an outdoor environment.
- Using a variety of devices, the children gathered information about the woodland habitats performing basic scientific enquiry and hypothesis testing.
- In a wood near Sussex University a group of school children gather for a day of activities. The children divide into pairs and each pair are given a PDA, a light and moisture probe and a small walkie-talkie so that they can communicate with a facilitator.

- The probe can be pointed into the air, placed into the ground, pressed against the bark of trees or even poked into noses - and produces a simple display of the light and moisture levels on its screen.
- Although the children do not realise this at the time, all the readings they make are being recorded through a wireless network and the probes are equipped with GPS so that the location of each reading is also recorded.
- As the children walk through the wood they pass near wireless beacons. Passing these may trigger messages and information to appear. Passing one beacon they hear a strange slurping sound - a massively amplified recording of a butterfly sucking nectar!
- At one point in the wood the children find a strange contraption shaped rather like an old hat-rack or a small metallic twiggy tree. This is the 'periscope'. It is used for seeing things that are normally unseen.
- The form of the device was designed by partners from the Royal College of Art trying to capture some of the sense of symbiosis between technology and the organic environment of the wood.
- Inside the hood of the periscope is a small computer screen and this is controlled by the 'handles' of the periscope. Moving the handles from side to side pans a virtual view of the wood. Twisting them makes the image move up and down. This is a rather odd mix of virtual and augmented realities.
- The periscope is set in the real environment and portrays a virtual view of the environment, but you can only see the real environment round the side of not through the image. Children would walk behind the periscope and expect that their friends could see them in the virtual image. Although our lives are filled with technology, we are still cognitively 'built' for the real world and find it hard to adjust to these half-real situations.

- The computers communicate via standard wireless networks (802.11b) as used in many offices, airports, etc. This is fairly standard technology, but the wood is not a standard environment: the natural environment of the wood is inimical to wireless networks.

- The wood seems a very open environment compared to the walls and ceilings of a building, but the leaves and wood of the trees are 90% water and water absorbs the radio waves.

- It is impossible to have just a few wireless base stations that talk to everything and all the computers had to be set up to relay wireless messages to one another. Even then they had to be placed below the leaf canopy as in summer the leaves blot out radio transmission at even very short range.

- Ambient Wood's technology doesn't just depend on computers, behind the scenes also a group of technicians were constantly working to keep the various aspects of the wood running, not least literally running around between outside sessions replacing the batteries in the many, many devices. The wood doesn't come equipped with mains electricity.

Q.4 Discuss about toward multi-scale and distributed output.

Ans. :

- The incorporation of ubiquitous computing capabilities into everyday life also requires novel output technologies and techniques. Output is no longer completely in the form of self-contained desktop/laptop visual displays that demand our attention.
- A variety of sizes or scales of visual displays, both smaller and larger than the desktop, are being distributed throughout the environments.
- The trend toward peripheral output has been explored for a particular class of displays, called ambient. Ambient displays need minimal attention and cognitive effort and are thus more easily integrated into a persistent physical space.

- One of the first ambient displays, the Dangling String. The Dangling String shares many features with successive efforts in ambient displays. A data resource drives the conceptual representation such that the output can be monitored by the user's peripheral perception.

Q.5 What do you mean seamless integration of physical and virtual worlds?

Ans. :

- An essential feature of ubicomp technology is that it attempts to combine computational artifacts smoothly with the world of physical artifacts.
- Researchers have suggested methods for using objects in the physical world to manipulate electronic objects, creating so-called graspable or tangible user interfaces.
- Sensors attached to devices themselves provide ways for physical manipulations of those devices to be interpreted appropriately by the applications running on those devices.

Q.6 What is context-aware computing?

Ans. :

- Location of identifiable entities (usually people) is a very regular piece of context used in ubicomp application development. The main general applications have been GPS-based car navigation systems and handheld 'tour guide' systems that differ the content displayed (video or audio) by a handheld unit given the user's physical location in an exhibit area.

- There is more to context than position (where) and identity (who). Although a full definition of context remains an exclusive research challenge, it is clear that in addition to who and where, context awareness involves :

1. **When** with the exception of with time as an index into a captured record or summarizing how long a person has been at a particular location, most context driven applications are unaware of the passage of time.

2. **What way of interpreting the 'what' of context is to see it as the focus of attention of one or more people during a live event.** Knowledge of the focus of attention at a live event can inform a better capture of that event.
3. **Why even more challenging than perceiving 'what' a person is doing is understanding 'why' they are doing it.** Sensing other ways of contextual data that could give an indication of a person's affective state.
- There are several of indoor positioning schemes as well, with conflicting characteristics in terms of cost, range, granularity and requirements for tagging and no particular solution is likely to ever meet all requirements

Q.7 Explain evaluation challenges for ubicomp.

Ans.:

- Formative and summative evaluation of ubicomp systems is hard and represents a real challenge for the ubicomp community.
- It is not clear that all the measures can apply universally across activities when we move away from structured and paid work to other activities.
- The shift away from the world of work means that there is still the question of how to apply qualitative or quantitative evaluation methods.
- The technology used to create ubicomp systems is often on the cutting edge and it is difficult to create reliable and robust systems that support some activity on a continuous basis.

6.2 Design Thinking

Q.8 What is design thinking?

Ans.: The design thinking process is nonlinear to users, to challenges assumptions, to redefine issues and to create innovative prototype and testing solutions. Design thinking is an iterative process. It is best to tackle problems that are not clearly defined or unknown through five phases - sympathise, define, ideate, prototype and test.

Q.9 Explain five stages of design thinking.

Ans.: Stages are as follows:

- **Stage 1:** Empathize - Research on users' needs. We should gain an insight into the problem, typically by researching users. Empathy, like design thinking, is crucial to a humane design process, because it enables own assumptions about the world to be set apart and real insight into users and their requirements.
- **Stage 2:** Define - State users' needs and problems: The time has come to collect information during the stage of empathy. We then analyse and synthesise observations to determine the key problems identified by us and our team. These terms are referred to as problem statements. We can build people to help keep your efforts personally focused before thinking.
- **Stage 3:** Ideate - Challenge assumptions and create ideas: We are now willing to generate thoughts. With a sound knowledge background in the first two phases, we can start thinking outside the box, find alternative ways to view the problem and find innovative solutions to the problem statement. Brainstorming here is especially helpful.
- **Stage 4:** Prototype - Start to create solutions: This is a testing phase. The objective is to determine the best solution possible for each problem. To examine the ideas generated, team should produce some cheap, scaled-down versions of the product. This might just involve prototyping paper.
- **Stage 5:** Test - Try solutions out assessors test the prototypes rigorously. Design thinking is iterative even if this is the final phase: Teams often redefine the results or more problems. We can therefore come back to earlier phases to make additional iterations, changes and refining - to find alternatives or to rule them out.

6.3 Finding Things on Web

Q.10 What is graphical user interface ?

Ans. :

- Graphical User Interface (GUI) is an interface or interactive system, that allows professionals to accomplish tasks on their computers through images and icons, rather than text command systems.
- Graphical user interfaces appear in computers, tablet devices and mobile devices. These graphical user interfaces can often be optimized to provide a more positive user experience.
- User interface is a collection of techniques and mechanisms to interact with somethings. In graphical interface, the primary interaction mechanism is a pointing device of same kind.
- User interacts with collection of elements referred to as objects. They can be seen, heard, touch.

Q.11 What is Hypertext ? Explain.

Ans. :

- Hypertext is text which contains links to other texts. The term was coined by Ted Nelson around 1965. HyperMedia is a term used for hypertext which is not constrained to be text: it can include graphics, video and sound. Apparently Ted Nelson was the first to use this term too.

- Hypertext consists of nodes connected by links to form networks or webs. Depending on the system, a node can be restricted to one medium (text, graphics, sound, animation or video) or can include multiple media.
- Links can be unidirectional or bidirectional, labeled or typed and can store other information, such as author and creation date.
- Anchors are points or regions in a node to which a link attaches, often represented by a button or other marking that indicates a navigational possibility.

- When a user navigates to a new node, a new window may open or the existing window may expand to incorporate the new information.

- Like hypertext, multimedia began with experiments in the 1960s and 1970s that matured into vigorous commercial activity in the 1980s and 1990s.

Q.12 List the main standard components of web.

Ans. :

- The main standard components of web :
 1. HyperText Markup Language (HTML)
 2. Uniform Resource Locators (URLs)
 3. HyperText Transfer Protocol (HTTP)

Q.13 Describe static web content.

Ans. :

- i. **The message and the medium** : On the web documents, users want to see the information and to retrieve it, which have an influence on design. The proper copy of information does not have the same inbuilt hypertext and active capabilities as the web page.

- ii. **Text** : The text information on web can be presented with different fonts, styles, colors, sizes in order to make the text more attractive and readable as per the importance of text in the context.

- iii. **Graphics** : There are number of sites on web that contain archives of graphical images, icons, backgrounds and so on. There are also paint and image manipulation packages available on almost all computer systems and scanners and digital cameras, where available, enable the input of photographs and diagrams.

- iv. **Movies and Sound** : Movies and Sound are both available to users of the web and hence to the page designers. It is made available with the help of different file formats.

Q.14 Discuss dynamic web content.**Ans. :**

i. **The active web :** This type involves complex forms of interactions on web. The actual content may be fixed, but the user can change the form of presentation. The web pages can be generated from database contents and the database information can be updated through the web.

ii. **Fixed content - local interaction and changing view :** Probably the most overestimated aspect of the web in recent years has been java. In fact, Java can be used to write server-end software and platform independent standalone programs.

iii. **Search :** The user's keywords are submitted to the server using an HTML form; they are compared against pre-prepared indexes at the server.

iv. **Dynamic content :** The content of the web pages reacts to and is updateable by the web user in dynamic web context. The user interacts through a web browser with a web server. The Java Servlet Pages (JSP) and Java Enterprise Beans (JEB) is used for this purpose. There is use of 'business logic' in data processing for e.g. online banking. These Java enterprise beans takes data from corporate database using JDBC connections.

Q.15 Explain characteristics and disadvantages of the graphical user interface.**Ans. : Characteristics :**

- A graphical system possesses a set of defining concepts. Included are sophisticated visual presentation, pick-and click interaction, a restricted set of interface options, visualization, object orientation, extensive use of a person's recognition memory and concurrent performance of functions.
- In a GUI multiple windows with different information can simultaneously be displayed on the user screen.

• Iconic information representation and symbolic information manipulation is possible in a GUI. Symbolic information manipulation such as dragging an icon representing a file to a trash can be deleting is intuitively very appealing and the user can instantly remember it.

• A GUI usually supports command selection using an attractive and user-friendly menu selection system.

• In a GUI, a pointing device such as a mouse or a light pen can be used for issuing commands. The use of a pointing device increases the efficacy issue procedure.

• Visualization is a cognitive process that allows people to understand information that is difficult to perceive, because it is either too voluminous or too abstract.

• Object orientation: A graphical system consists of objects and actions. Objects are what people see on the screen as a single unit.

• Pick-and-click interaction : The primary mechanism for performing this pick-and-click is most often the mouse and its buttons. The user moves the mouse pointer to the relevant element (pick) and the action is signaled (click). Pointing allows rapid selection and feedback.

• Actions : A series of actions may be performed on a selected object. Performing a series of actions on an object also permits and encourages system learning through exploration

Disadvantages :

1. Greater design complexity
2. May consume more screen space
3. Good design also requires hardware of adequate power, processing speed, screen resolution and graphic capability.
4. Inefficient for touch typists: For an experienced touch typist, the keyboard is a very fast and powerful device.

5. Not always the preferred style of interaction: Not all users prefer a pure iconic interface. User will also prefer alternatives with textual captions.

6. Not always fastest style of interaction : Graphic instructions on an automated bank teller machine were inferior to textual instructions.

Q.16 Why graphics is popular ?

Ans. :

- A graphical screen bore scant resemblance to its earlier text-based colleagues. Older text-based screen possessed a one dimensional.
- Graphic screens assumed a three-dimensional look. Controls appeared to rise above the screen and move when activated. Information could appear and disappear, as needed.
- Text could be replaced by graphical images called icons. These icons could represent objects or actions.
- Selection fields such as radio buttons, check boxes, list boxes and palettes coexisted with the reliable old text entry field.
- Objects and actions were selected through use of pointing mechanisms. It increased computer power. User's actions to be reacted to quickly, dynamically and meaningfully.
- Graphic presentation is much more effective than other presentation methods. Properly used, it reduces the requirement for perceptual and mental information recoding and reorganization and also reduces the memory loads.
- It permits faster information transfer between computers and people by permitting more visual comparisons of amounts, trends or relationships; more compact representation of information.
- Graphics also can add appeal or charm to the interface and permit greater customization to create a unique corporate or organization style.

6.4 Augmented Reality, Virtual Reality

Q.17 What is augmented reality ?

Ans. :

- Augmented Reality (AR) is an enhanced version of the real physical world achieved by means of technology-driven digital visual elements, sound or other sensory stimuli. The trend among mobile computers and business applications in particular is growing.
- With the increasing collection and analysis of data, one of the primary purposes of increased reality is to highlight certain characteristics of the physical world, improve its realisation and gain intelligent and accessible insight that can be applied to real world applications. Such large data can inform decision-making by enterprises and gain insights into, inter alia, consumption habits.

Q.18 Define virtual reality.

Ans. : Virtual reality is a high-end user-computer interface that involves real time simulation and interactions through multiple sensorial channels. These sensorial modalities are visual, auditory, tactile, smell and taste.

Q.19 Explain about virtual reality on the desktop and in the home.

Ans. :

- Virtual reality has been made possible by the advent of very fast high-performance computers. Despite the exponential rise in processor speeds, high-resolution immersive VR is still not available for mass-market applications and many systems are primarily research projects.
- Desktop virtual reality is a lower-cost alternative. In desktop virtual reality, 3D images are presented on a normal computer screen and manipulated using mouse and keyboard, rather than using goggles and data-gloves.

- Many readers may have used such systems on personal computers or games consoles: flight simulators or interactive games such as DOOM.
- Virtual Reality Markup Language (VRML) can include static 3D objects, which the user can navigate around looking at different aspects and dynamic objects that move about and react when 'touched' by the mouse cursor.
- Internet-based VR systems are also available that allow greater interactivity and collaboration with other remote users.

6.5 Challenges in Designing Interfaces for Smart Homes

Q.20 What is a smart home ?
Ans. :

- A smart home is a house which is equipped with connected devices that can be programmed and controlled remotely via a smartphone or computer. For example, a smart home enables remote control of lighting, temperature, multi-media, security, window and doors and many other functions.
- The most common smart devices include : Lighting, Heating, Power controls (switches and smart plugs), Energy monitors, Security and Home entertainment systems.
- Smart homes use home automation technologies to provide home owners with feedback and information by monitoring aspects of the home and providing smart, programmable controls.
- For example, a smart refrigerator may be able to catalogue its contents, suggest menus, recommend healthy alternatives and order replacements as food is used up. Smart devices can even take care of feeding the cat and watering the plants, even when the owner is not at home.

Q.21 Explain about home automation.

Ans. :

- Home automation is the automatic control of electronic devices in your home. These devices are connected to the Internet, which allows them to be controlled remotely.
- Interconnected devices enable to intelligently monitor and control smart homes in a future Internet of Things.
- Energy saving applications, for example, control indoor climate and electricity usage by employing context information to switch off appliances (e.g., lights, computers), reduce room temperature, close windows or stop warm water circulation.
- Home automation works on three levels :

1. **Monitoring :** Monitoring means that users can check in on their devices remotely through an app. For example, someone could view their live feed from a smart security camera.
2. **Control :** Control means that the user can control these devices remotely, like planning a security camera to see more of a living space.
3. **Automation :** Finally, automation means setting up devices to trigger one another, like having a smart siren go off whenever an armed security camera detects motion.

Q.22 Explain benefits of smart home technology.

Ans. : Benefits are as follows :

- **Remote monitoring :** Smart devices enable real-time monitoring when we are away from home, providing useful data and increased security.
- **Interconnectivity :** Linking devices can improve their functionality and enable them to work together to deliver coordinated results.
- **Safety :** Smart devices can trigger alerts when certain events happen, improving the safety of the home.

- **Fault detection** : Smart devices can monitor home for leaks and flooding and provide real-time alerts to avoid costly damage.
- **Motion detection** : By activating only when motion is detected, this technology saves time, battery life, memory, storage and energy.
- **Security** : By linking motion detection, cameras, alarms and security services smart technology provides increased home security.
- **Energy saving** : Smart devices can educate users about the home's energy consumption and production from renewables, helping to increase efficiency, reduce carbon emissions and costs.

Q.23 What are the challenges in designing interfaces for smart homes?

Ans. : Challenges

- **Does your smart home understand you?** : It must still be added that, while communication to people is a natural activity, adaptation is slow. Even though the early adopters with Google or Echo devices have built their homes and could not imagine life without a smart kitchen, smart safety and intelligent climate control, the majority of them, including me, still get used to how voice is integrated in our homes.

- **What to look out for as visual designers?** : This is a great opportunity for us, designers, to transition from screen to voice and to start creating products that are much easier to use, to make people's lives easier by changing the way they perform everyday tasks. Not to mention spending more time with other people rather than interacting with all sorts of devices. So how can we design smart products that improve users' wellbeing at home? And what additional competencies should we have in order to keep up with the evolving tech?

- For starters, we have to find out the users' needs and give them a solution within the overall design. Think of it this way: the user has a problem, what kind of solutions can we create to solve it?

Human Computer Interaction

- Use cases can help define the ways users might interact with a smart device.
- Besides the usual team, other professionals such as electricians or architects might come into the picture whom you should consult the feasibility of the product within the home.
- It should be an iterative process, but it should be considered that changes on a physical object might be more expensive, that's why the discovery phase should be very thorough.
- Defining the personas is a good way to find out the tone of voice of the product.

- **The best smart home devices will hear what people say** : But there are multiple conditions that have to be considered when defining the way people will speak to the device. As mentioned earlier, the users might have an accent or speak from far away. They might get interrupted or might cough during a command. But there could be many other similar factors that can challenge the device's understanding of what the users meant. Not to mention the countless ways they can express one command. There could be a very technical persona who would tell a command in great detail whereas another could be more vague. The device should be able to execute the action even if the user didn't give all the information that was needed to do so. It has to draw a conclusion from the context. Try to understand the intent then confirm it with a number of questions. So instead of making the user repeat the same command, make the device ask for the missing information instead. This will result in a more natural conversation. Also, the AI should let the user know that the command has been confirmed and executed.

- **Thinking like a user is key for a smooth experience** : It is very important to define how they would like to get from A to B. Building up as many flows as possible of the whole process between the user and a device takes time and it shouldn't be

rushed. The device will fail multiple times but that's totally okay, this is the way it will be improved. So when defining the structure of speech, the main things to consider are the following : what people say, what they actually mean and what the result should be. After defining the persona and all their potential ways to communicate, the device's persona should be tailored accordingly. Is it friendly ? Does it match the company brand ? Does it sound pleasant ? Moving on with the competences, designers should also have a solid knowledge of system design and awareness of consistency. If the designed product has a built-in Alexa for example, it should follow the guidelines to keep consistent experience with what people are already used to. Unless user tests prove otherwise. Even though the main interaction with smart devices will be based on voice, they need to have a screen or some sort of visual indicators. The best way to decide whether the device needs a screen or if just a simple LED light will be enough, is to test it with users.

- **Communication should feel natural with smart home products** : People are used to getting feedback, seeing a reaction when talking to someone. Thus smart devices need to show somehow that they can hear the user or indicate that they are waiting for a command or processing it. Another important factor is being able to turn off the device's microphone, as most people are very concerned about their privacy. They still feel uncomfortable by having a device in the house that hears everything. This might be one of the reasons why adaptation is so slow. So having an indicator that shows when the device is turned on is one of the most important features. If the product has a screen, it is important to show the main functions of the device rather than the built-in voice AI's functionality. Another important factor is to separate the visual controls from the voice experience. Also, keep in mind that people might stand far from the device so the elements on the

screen should be of a size that can be easily viewed from a distance. This means that it shouldn't be overcrowded. It should be clean and to the point containing only the necessary elements. Here's an interesting case study about Google Home Hub's visual experience.

6.6 Smart Device and Handheld Device

Q.24 What is smart device ? Explain its characteristics.

Ans. :

- An smart device is an electronic device, which usually can operate interactively and autonomously with other devices or network systems through various wireless protocols such as Bluetooth, Zigbee, NFC, WiFi, LiFi, 5G, etc.
- Smart phones, intelligent vehicles, smart thermostats, smart doorbells, intelligent locks, smart fridges, phablets and tablets, clever smart bands, smart key chains, false and many others are noticeable types of smart devices.
- This term could also mean a device which has certain characteristics of omni-present computing, including artificial intelligence, although not necessarily.
- Smart devices typically consist of a hardware layer (including a signalling radio), a network layer (through which systems interact) and a use layer (through which end users deliver commands).
- It has following characteristics :
 1. A set of system hardware and software IT resources. This set is usually static fixed at design time.
 2. Dynamic component-oriented resource extensions and plug-ins (Plug and play) of some hardware resources.
 3. Remote external service access and execution.
 4. Local, internal autonomous service execution.

5. Access to specific external environments: human interaction, physical world interaction and distributed ICT / virtual computing interaction.

• Common types of smart devices include :

1. Tab and pad type smart devices that often act as personalised smart mobile devices.
2. Smart environment devices.

Q.25 What are handheld devices ?

Ans. : Handheld device is a pocket-sized computing device with a display screen and input/output interface like an external or touch screen keyboard. Going by this definition of handheld devices and gadgets many appliances can qualify to be called handheld, like a mobile phone, PDA, mobile PC, handheld game consoles and so on.

Q.26 Discuss about handheld device format apps.

Ans. :

- Mobile apps are very different; the width and height of elements are set by points. The layout does have the ability to wrap, but not re - stack or re - arrange unless it's specifically coded to do so for a particular device size and that can lead to way too much development effort.
- Modern handheld - format devices also continue to use some of the same basic layout patterns employed in these early systems.
- 1. **Stack**
 - Primary pattern used by non - game mobile application is stack. It is handheld device. Narrow and tall form factor of smartphones and other handheld mobile device dictates a list - like display for most types of content or control.
 - Fig. Q.26.1 shows mobile apps use a stack layout pattern including content, control and navigation elements.

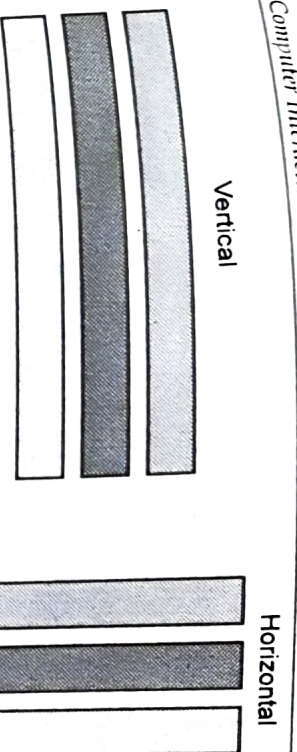


Fig. Q.26.1 : Stack layout

- Stacks are vertically organized structures with a content area, usually arranged in a list or grid. Most iOS, Android and Windows Phone apps follow this top - level pattern.
- Stack layout positions child elements in a stack either horizontally or vertically based on the orientation property.

2. Screen carousels

- Alternative to top - level pattern and used in dashboard like display. It has multiple instances or variants between which the user can quickly navigate via a swipe gesture to the left or right. The iOS weather app is an example of screen carousels.
- Carousels may or may not have top or bottom bars associated with them, but they usually do have a page marker widget that shows the user's place in the carousel content.
- Carousels don't provide circular flow, but rather disallow further swiping at the far left and right.
- 3. **Orientation and layout**
 - Latest mobile devices can detect their screen orientation i.e. portrait or landscape. This helps the app to rearrange its layout to better suit the current orientation. The majority of apps stick with portrait orientation even when rotated.
 - For list - and grid - based content browsing assuming portrait orientation is a good because users usually operate the phone one - handed in portrait orientation. For applications such as photo or video capture and editing, it makes sense to allow rotation to landscape orientation, since the medium itself can be in that orientation.

Q.27 List and explain challenges in HCI design for mobile devices.

Ans. : Challenges in HCI design for mobile devices are of hardware and software.

Hardware challenges

- Due to the limitations of size and weight for portability purpose, the interface design for mobile devices comes with more hardware challenges.
- These challenges include limited input facilities, limited output facilities and designing for mobility.
- There are 3 main inputs in current mobile device : Keyboard, Stylus on the touch screens and Scroll wheel
- The keyboard allows a user to hit a key to perform a task or navigate through the mobile menu functionalities, the stylus with the touch screen allows a user to hit the screen to do the task; the scroll wheel can be scrolled and pushed by a user to do a task and also navigate through the menus and submenus.
- The design of keyboards for mobile devices has been a challenge because the space for key installation on a mobile device is limited.
- Touch input would be problematic if the screen of a mobile device is small and that would lead a user's fingers to occlude the graphical elements he wishes to work with.
- Scroll wheel can be used to navigate a mobile device menu in one direction, either horizontally or vertically. It can also be used as a push button to do a specific task to support the use of one hand to interact with the mobile device.
- Limited output facilities : The small-sized screen is one of the mainly and most commonly used output facilities for mobile devices. Designing the screen for outputting is a trade-off challenge that needs to be experimentally studied to find out which is the efficient and most effective size of the screen that can be used for the different types of mobile devices.

DECODE

Software challenges

- System of menus : Taking a successful design from a desktop and apply it to a mobile device without a clear understanding of the translation inputs and outputs can lead to an ineffective interaction design.
- The mainly and widely used alternative is the use of hierarchical menus. With a hierarchical menu, a user can select a menu item that can then open another submenu; and so on until the user reaches the desired function he or she is aiming to reach.
- Navigating and browsing: to display information that is well suited for larger screens, the information has to be segmented into many small presentation units that can fit into the small screen of mobile devices. And this makes it difficult to effectively organize information and help users navigate to and from the information they want.
- Images and Icon: display of graphical content described by raster and vector graphics on mobile devices to allow appropriate and resource-saving implementations.

Q.28 How to choose breakpoints ?

Ans. :

- Do not define breakpoints based on device classes.
- Defining breakpoints based on specific devices, products, brand names or operating systems that are in use today can result in a maintenance nightmare. Instead, the content itself should determine how the layout adjusts to its container.
- Create breakpoints based on content, never on specific devices, products or brands.
- Design for the smallest mobile device first; then progressively enhance the experience as more screen real estate becomes available.
- Keep lines of text to a maximum of around 70 or 80 characters.

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- Pick major breakpoints by starting small.
- Design the content to fit on a small screen size first, then expand the screen until a breakpoint becomes necessary. This allows you to optimize breakpoints based on content and maintain the least number of breakpoints possible.
- Pick minor breakpoints when necessary.

6.7 Smart Wrist Watch, Future of HCI

Q.29 Write short note on smart wrist.

Ans. :

- A smartwatch is a wearable computer in the form of a watch; modern smartwatches provide a local touch screen interface for daily use, while an associated smartphone app provides for management and telemetry.
- While early models could perform basic tasks, such as calculations, digital time telling, translations and game-playing, 2010s smartwatches have more general functionality closer to smartphones, including mobile apps, a mobile operating system and WiFi / Bluetooth connectivity.
- Some smartwatches function as portable media players, with FM radio and playback of digital audio and video files via a Bluetooth headset. Some models, called watch phones, have mobile cellular functionality like making calls.
- While internal hardware varies, most have an electronic visual display, either backlit LCD or OLED. Some use transreflective or electronic paper, to consume less power. They are generally powered by a rechargeable lithium-ion battery.
- Peripheral devices may include digital cameras, thermometers, accelerometers, pedometers, heart rate monitors, altimeters, barometers, compasses, GPS receivers, tiny speakers and micro SD cards, which are recognized as storage devices by many other kinds of computers.

- Software may include digital maps, schedulers and personal organizers, calculators and various kinds of watch faces. The watch may communicate with external devices such as sensors, wireless headsets or a heads-up display.
- Like other computers, a smartwatch may collect information from internal or external sensors and it may control or retrieve data from, other instruments or computers. It may support wireless technologies such as Bluetooth, Wi-Fi and GPS.
- For many purposes, a "watch computer" serves as a front end for a remote system such as a smartphone, communicating with the smartphone using various wireless technologies. Smartwatches are advancing, especially their design, battery capacity and health-related applications. Health-related applications include applications measuring heart rate, SpO2, workout etc.

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